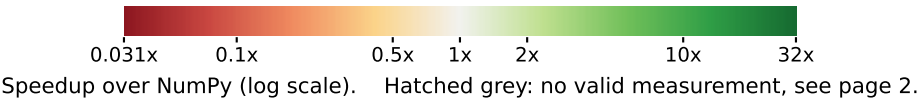


DaCe auto_optimize on the non-PolyBench NPBench kernels

Median speedup over NumPy (50 repetitions)

	Pluto	DaCe auto_opt	NumPy	
geo-mean				
cholesky2	n/a	invalid		PolyBench family - no Pluto reference yet
covariance	n/a	invalid		
covariance2	n/a	invalid		
doitgen	n/a	invalid		
gesummv	n/a	invalid		
jacobi_1d	n/a	invalid		
k2mm	n/a	invalid		
k3mm	n/a	invalid		
symm	n/a	invalid		
conv2d_bias	n/a	invalid		deep learning
lenet	n/a	invalid		
mlp	n/a	invalid		
resnet	n/a	invalid		
softmax	n/a	invalid		
hdiff	n/a	invalid		weather stencils
vadv	n/a	invalid		
cavity_flow	n/a	invalid		physics / CFD
channel_flow	n/a	invalid		
contour_integral	n/a	invalid		
nbody	n/a	invalid		
scattering_self_energies	n/a	invalid		
arc_distance	n/a	invalid		signal / integration
azimint_hist	n/a	invalid		
azimint_naive	n/a	invalid		
compute	n/a	invalid		
stockham_fft	n/a	invalid		
crc16	n/a	invalid		misc / non-affine
go_fast	n/a	invalid		
mandelbrot1	n/a	invalid		
mandelbrot2	n/a	invalid		
spmv	n/a	invalid		



DaCe auto_optimize on the non-PolyBench NPBench kernels: detailed results

Preset paper, 50 repetitions per kernel. Every blank cell on page 1 is accounted for below; a speedup is given only where the result validated.

DaCe column: dace_cpu_autoopt / b27aaed2f

Pluto: polycr --tile --parallel --codegen-context=1 (Clan frontend), clang -O3 -march=native

kernel	NumPy runtime	Pluto	DaCe auto_opt	validated	note / reason a cell is blank
cholesky2		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
covariance		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
covariance2		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
doitgen		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
gesummv		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
jacobi_1d		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
k2mm		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
k3mm		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
symm		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
conv2d_bias		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
lenet		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
mlp		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
resnet		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
softmax		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
hdiff		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
vadv		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
cavity_flow		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
channel_flow		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
contour_integral		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
nbody		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
scattering_self_energies		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
arc_distance		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
azimint_hist		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
azimint_naive		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
compute		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
stockham_fft		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
crc16		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
go_fast		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
mandelbrot1		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
mandelbrot2		n/a	invalid	no	Pluto: no result recorded DaCe: invalid
spmv		n/a	invalid	no	Pluto: no result recorded DaCe: invalid